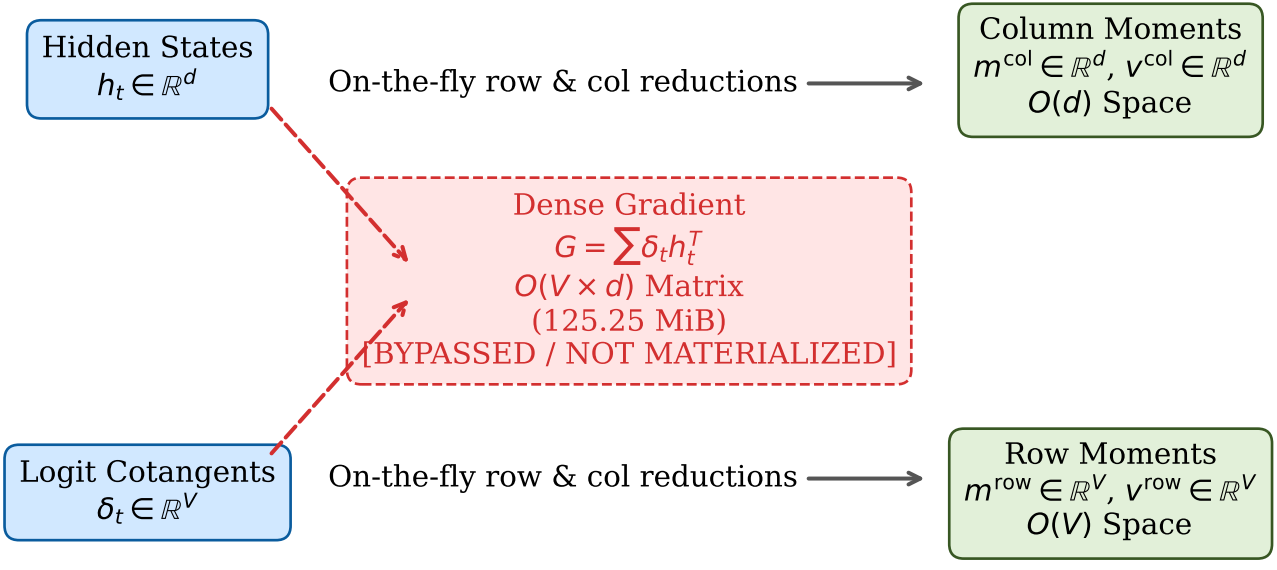
Hidden States
$h_t \in \mathbb{R}^d$
On-the-fly row & col reductions
Column Moments
$m^{\text{col}} \in \mathbb{R}^d$, $v^{\text{col}} \in \mathbb{R}^d$
$O(d)$ Space
Dense Gradient
$G = \sum \delta_t h_t^T$
$O(V \times d)$ Matrix
(125.25 MiB)
[BYPASSED / NOT MATERIALIZED]
Logit Cotangents
$\delta_t \in \mathbb{R}^V$
On-the-fly row & col reductions
Row Moments
$m^{\text{row}} \in \mathbb{R}^V$, $v^{\text{row}} \in \mathbb{R}^V$
$O(V)$ Space